

ALMATY, Kazakhstan

WMO: 368700

Lat: 43.2394N

Lon: 76.9328E

Elev: 2792

StdP: 13.27

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.6	1.7	-6.2	4.4	-2.4	-1.9	5.5	2.4	14.3	29.8	11.8	28.0	2.4	170	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	93.9	65.4	91.1	64.6	88.1	64.0	68.9	86.2	67.2	84.8	65.8	83.7	5.0	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.0	95.5	75.7	61.1	89.2	74.6	59.3	83.5	73.4	34.9	86.7	33.4	84.7	32.2	84.2	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
13.9	11.3	9.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-8.1	99.8	7.7	3.4	-13.7	102.3	-18.2	104.3	-22.5	106.2	-28.2	108.6
			-8.4	72.5	7.8	3.3	-14.0	74.9	-18.5	76.9	-22.9	78.7	-28.5	81.1
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.4	22.5	26.4	40.6	53.1	62.5	71.1	75.3	73.5	64.0	51.1	37.4	26.2	(6)
(7)		DBStd	20.33	9.92	10.02	10.57	8.65	6.81	5.76	5.31	6.06	7.77	8.51	9.69	9.29	(7)
(8)		HDD50	3120	853	660	318	66	4	0	0	3	90	388	738		(8)
(9)		HDD65	6250	1318	1079	755	364	128	15	2	112	437	830	1203		(9)
(10)		CDD50	3277	0	0	28	158	391	633	783	730	422	123	9	0	(10)
(11)		CDD65	932	0	0	0	6	50	198	320	272	81	5	0	0	(11)
(12)		CDH74	10349	0	0	3	106	524	2078	3587	3049	926	75	1	0	(12)
(13)		CDH80	4089	0	0	0	18	116	748	1574	1327	295	11	0	0	(13)
(14)	Wind	WSAvg	3.7	2.9	3.2	3.8	4.4	4.4	4.2	4.2	3.9	3.7	3.2	3.2	2.9	(14)
(15)	Precipitation	PrecAvg	25.6	1.3	1.5	2.8	4.1	4.1	2.4	1.5	1.1	1.1	2.2	2.2	1.5	(15)
(16)		PrecMax	39.9	2.8	2.7	6.1	8.7	8.4	7.6	5.0	2.9	3.9	5.8	5.0	3.3	(16)
(17)		PrecMin	14.1	0.3	0.2	1.1	0.0	0.9	0.3	0.1	0.0	0.0	0.2	0.3	0.2	(17)
(18)		PrecStd	4.8	0.6	0.6	1.3	2.1	1.8	1.3	1.1	0.8	0.8	1.4	1.0	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.9	53.7	71.4	82.4	87.9	93.6	98.4	97.1	91.2	81.0	68.0	49.8	(19)
(20)		MCWB	38.1	43.6	52.5	58.5	63.3	64.7	67.7	66.1	62.5	57.1	50.7	40.8		(20)
(21)		2%	DB	41.9	47.6	66.3	76.9	82.8	89.9	94.6	93.6	86.7	75.1	61.4	44.9	(21)
(22)		MCWB	35.6	40.2	50.3	57.3	62.5	64.9	65.7	65.2	61.2	55.2	48.7	38.3		(22)
(23)		5%	DB	38.4	43.1	61.6	72.4	79.8	87.5	91.4	90.8	83.3	70.2	55.5	41.1	(23)
(24)		MCWB	33.8	37.3	48.6	55.2	61.3	64.4	65.2	64.1	59.7	53.0	46.2	36.0		(24)
(25)		10%	DB	35.3	39.2	56.8	68.2	76.6	84.4	88.1	87.6	79.5	65.9	51.3	37.8	(25)
(26)		MCWB	32.2	34.9	46.7	53.5	59.8	63.5	64.6	63.4	58.3	51.5	44.1	34.2		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.6	44.2	54.7	62.7	66.6	70.1	71.9	71.1	66.2	58.5	53.6	42.9	(27)
(28)		MDCB	43.9	50.6	66.9	76.9	81.4	85.9	88.8	88.1	87.4	77.3	62.5	47.5		(28)
(29)		2%	WB	36.6	40.7	51.8	59.4	64.4	67.8	69.2	67.8	62.7	56.2	49.4	39.0	(29)
(30)		MDCB	40.8	46.7	63.8	71.7	78.9	83.8	87.1	86.5	82.7	71.5	58.7	43.4		(30)
(31)		5%	WB	34.3	37.7	49.2	56.6	62.6	66.1	67.5	66.1	60.8	54.4	46.9	36.5	(31)
(32)		MDCB	37.4	42.4	59.7	69.3	77.1	82.9	85.1	85.7	79.8	67.4	54.3	40.5		(32)
(33)		10%	WB	32.1	35.2	47.1	54.6	60.9	64.7	66.1	64.6	59.1	52.5	44.4	34.2	(33)
(34)		MDCB	35.0	38.5	56.1	65.9	74.1	81.0	84.1	83.7	77.5	64.0	50.4	37.4		(34)
(35)	Mean Daily Temperature Range		MDBR	17.5	17.4	18.6	20.6	21.4	22.6	23.2	24.4	24.1	22.1	17.3	16.2	(35)
(36)		5% DB	MCDBR	20.8	21.9	26.6	28.2	26.4	26.8	27.5	28.2	28.7	29.1	25.9	20.0	(36)
(37)		MCWBR	15.1	15.2	14.5	14.2	13.1	11.8	10.9	11.2	12.3	13.9	15.5	14.3	(37)	
(38)		5% WB	MCDBR	18.7	19.9	24.0	25.0	24.3	24.0	24.6	26.9	26.5	25.6	23.4	18.6	(38)
(39)		MCWBR	13.8	14.1	13.7	13.5	13.0	11.9	11.4	11.8	12.3	12.9	14.6	13.5		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.318	0.362	0.422	0.459	0.435	0.437	0.431	0.404	0.388	0.391	0.357	0.312	(40)	
(41)		taud	2.163	2.055	1.969	1.955	2.103	2.158	2.201	2.268	2.253	2.132	2.139	2.188	(41)	
(42)		Ebn at Noon	250	256	254	256	266	266	266	270	265	243	231	239	(42)	
(43)		Edh at Noon	31	41	51	56	49	47	44	40	38	37	31	27	(43)	
(44)	All-Sky Solar Radiation	RadAvg	640	922	1213	1564	1955	2075	2056	1919	1543	1013	625	500	(44)	
(45)		RadStd	48	75	93	133	130	84	122	94	82	87	61	41	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+1.64	+4.19	N/A	N/A	-1.58	-2.52	-365	-454	+250	+130	(46)
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	-0.94	-1.60	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page